Chapter 1. Pits, passes and peaks: critical points and the lemma of Morse

Milnor, "Morse Theory", §1–§2, rewritten for a first-year graduate student who wants to know what the theorem is for. Every proof is complete, except that the existence and smooth-dependence theorem for ordinary differential equations is quoted, as Milnor quotes it.

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0. In how many ways can a double pendulum fall?

A double pendulum is a rod hinged at a fixed pivot, with a second rod hinged to its free end, each rod carrying a mass at its tip, the whole free to swing in a vertical plane. Its position is given by two angles θ_1, θ_2, each measured from the downward vertical, so the space of all positions is a torus $S^1 \times S^1$. There are exactly four positions in which it can rest: both rods down, both rods up, and the two mixed positions. Only the first is stable, but the other three are unstable in different ways. Balanced with the inner rod down and the outer rod up, the pendulum has essentially one way to fall: the outer rod tips over, to one side or the other. Balanced with both rods up, it has two independent ways to fall.

The number of independent ways to fall is what Morse called the *index* of the equilibrium. It can be read off from the second derivatives of the potential energy $V(\theta_1, \theta_2)$: it is the number of negative eigenvalues of the Hessian matrix. The point of this chapter is that, as long as no eigenvalue is zero, the index tells the whole local story. The lemma of Morse says that in suitable curvilinear coordinates y^1, y^2 centred at the equilibrium,

$$V = V(\text{equilibrium}) \pm (y^1)^2 \pm (y^2)^2$$

exactly, not up to higher-order terms, with as many minus signs as the index. Near a non-degenerate equilibrium nothing survives a change of coordinates except the signs.

This is the local half of Morse theory. The global half (Chapter 2) says that the same critical points control how the regions $\{V \leq a\}$ change as a grows, and through them the shape of the whole space. Milnor explains both halves with a torus standing on a table, whose height function has one minimum, two saddles and one maximum. So does the double pendulum: the same picture, different physics.

Milnor's §2 ends with a more technical tool, the flow of a vector field (a one-parameter group of diffeomorphisms). It is what will slide one region $\{f \leq b\}$ down onto another in Chapter 2. We prove his Lemma 2.4 completely, quoting only the existence and smooth-dependence theorem for differential equations, which we state precisely.

The application at the end computes the four indices of the double pendulum, and the frequencies and growth rates of its small motions near each equilibrium. The index turns out to be the number of exponentially growing modes, which chemists call imaginary frequencies; a transition state of a chemical reaction is, by definition, an equilibrium of index one.

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1. A torus standing on a table (Milnor pp. 1–3)

Stand a torus M upright on a horizontal plane V, touching it at one point, and let $f : M \to \mathbb{R}$ be the height above V ($\mathbb{R}$ always denotes the real numbers). For $a \in \mathbb{R}$ put

$$M^a = f^{-1}(-\infty, a] = \{x \in M : f(x) \le a\},$$

the part of the torus that would be under water if the room were flooded to depth a. There are four points where the tangent plane is horizontal; call them p, q, r, s from the bottom up. Milnor reads the following off the picture:

1. if $a < 0 = f(p)$, then M^a is empty;
2. if $f(p) < a < f(q)$, then M^a is homeomorphic to a 2-cell (a closed disk);
3. if $f(q) < a < f(r)$, then M^a is homeomorphic to a cylinder $S^1 \times [0,1]$;
4. if $f(r) < a < f(s)$, then M^a is homeomorphic to a compact surface of genus one with a circle as boundary;
5. if $f(s) < a$, then M^a is the whole torus.

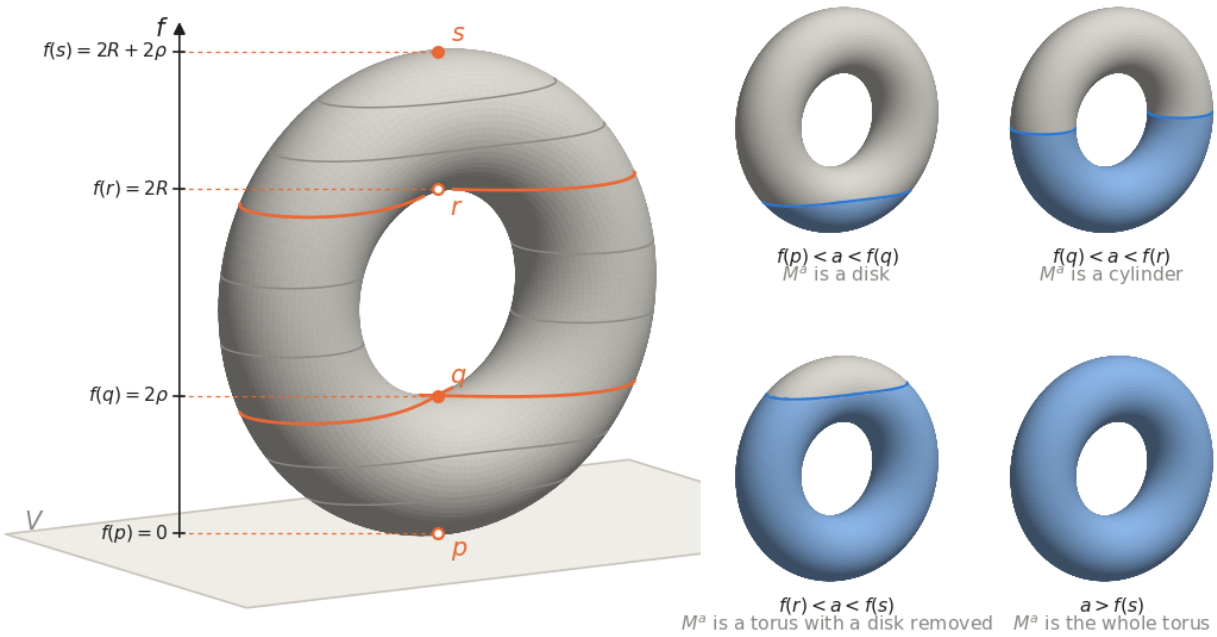
To have something to compute with, take the torus of revolution with tube radius ρ and core radius $R > \rho$, with its axis horizontal at height $R + \rho$:

$$(u, v) \longmapsto \big((R + \rho\cos v)\cos u,\ \rho \sin v,\ R + \rho + (R + \rho\cos v)\sin u\big), \qquad u, v \in \mathbb{R}/2\pi\mathbb{Z}.$$

In the angular coordinates, $f(u, v) = R + \rho + (R + \rho\cos v)\sin u$, and p, q, r, s are $(u, v) = (-\frac{\pi}{2}, 0),\ (-\frac{\pi}{2}, \pi),\ (\frac{\pi}{2}, \pi),\ (\frac{\pi}{2}, 0)$, at heights $0,\ 2\rho,\ 2R,\ 2R + 2\rho$. (That these are the only critical points is checked in §3.)

The figure is Milnor's Diagram 1 for this torus, with $R = 2.5\rho$. On the left, the torus stands on V, and level curves $f = a$ are drawn at several heights. At $a = f(q)$ and $a = f(r)$ the level curve is a figure eight crossing itself at q, resp. r (in colour; its back half is hidden). A hollow dot is a critical point on the far side. On the right, M^a is shaded as water for one a in each of the ranges (2)–(5). In the web edition the water level a is a slider, and the figure says what M^a is at each height.

(In the printed version this figure is a still image; the interactive version is in the web edition.)



As a increases the homeomorphism type of M^a jumps in complicated ways; its *homotopy type* changes in a simple way. Recall that a continuous map $h : X \to Y$ is a *homotopy equivalence* if there is a continuous $k : Y \to X$ with $k \circ h$ and $h \circ k$ homotopic to the identity maps; X and Y then have the *same homotopy type*, written $X \simeq Y$. In these terms:

- (1) $\to$ (2) is the operation of attaching a 0-cell: as far as homotopy type is concerned, the disk M^a, $f(p) < a < f(q)$, cannot be distinguished from a point;
- (2) $\to$ (3) is attaching a 1-cell: the cylinder $\simeq$ the disk with an arc attached at its two ends;
- (3) $\to$ (4) is again attaching a 1-cell;
- (4) $\to$ (5) is attaching a 2-cell.

Definition (attaching a cell). Let $e^k = \{x \in \mathbb{R}^k : \|x\| \le 1\}$, the *k-cell*, and let $\dot{e}^k = \{x \in \mathbb{R}^k : \|x\| = 1\}$, its boundary, be denoted S^{k-1}. If Y is a topological space and $g : S^{k-1} \to Y$ is continuous, then $Y \cup_g e^k$, *Y with a k-cell attached by g*, is the quotient of the disjoint union $Y \sqcup e^k$ by the equivalence relation generated by $x \sim g(x)$ for $x \in S^{k-1}$. For $k = 0$, e^0 is a point and $\dot{e}^0 = S^{-1} = \emptyset$, so Y with a 0-cell attached is Y together with a disjoint point.

The points p, q, r, s at which the homotopy type of M^a changes are the *critical points* of f: in any coordinates (x, y) near them, $\partial f/\partial x = \partial f/\partial y = 0$. Moreover there are coordinates in which

$$f = x^2 + y^2 \text{ near } p, \qquad f = \text{const} + x^2 - y^2 \text{ near } q \text{ and } r, \qquad f = \text{const} - x^2 - y^2 \text{ near } s.$$

The number of minus signs at each point is the dimension of the cell that must be attached to pass from M^a to M^b when $a < f(\text{point}) < b$. Milnor's first theorems generalize this to any smooth function on any manifold. The normal forms are the lemma of Morse (§4 below); the cell attachments are Theorems 3.1 and 3.2 of Chapter 2, where the homotopy-theoretic content of (1)–(5) is proved rather than read off a picture.

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2. Critical points and sublevel sets (Milnor p. 4)

"Smooth" and "differentiable" both mean C^∞. A *manifold* is a smooth manifold of dimension n, without boundary, whose topology is Hausdorff and has a countable basis. Milnor does not list these conditions; we point out where each is used. The tangent space of M at p is TM_p, and a smooth $g : M \to N$ with $g(p) = q$ induces the linear map $g_* : TM_p \to TN_q$.

Definition. Let $f : M \to \mathbb{R}$ be smooth. A point $p \in M$ is a *critical point* of f if $f_* : TM_p \to T\mathbb{R}_{f(p)}$ is zero; $f(p)$ is then a *critical value*. Any other real number, including one that is not a value of f, is a *regular value*.

In a coordinate system $(x^1, \dots, x^n)$ on a neighbourhood U of p, f_* sends $\partial/\partial x^i|_p$ to $\frac{\partial f}{\partial x^i}(p)\,\frac{d}{dt}$, so p is critical if and only if

$$\frac{\partial f}{\partial x^1}(p) = \cdots = \frac{\partial f}{\partial x^n}(p) = 0.$$

This agrees with the definition in the companion volume (*Topology from the Differentiable Viewpoint*, rewritten; Chapter 2, §1), where a critical point is one at which the derivative is not surjective: a linear map to $\mathbb{R}$ is surjective exactly when it is nonzero. That volume works with manifolds lying in some $\mathbb{R}^k$; every result we cite from it is local and carries over to the abstract manifolds used here by working in a chart (for Proposition A the chart argument is written out below).

Proposition A (Milnor: "it follows from the implicit function theorem"). If a is a regular value of f, then M^a is a smooth n-manifold with boundary $f^{-1}(a)$, and $f^{-1}(a)$ is a smooth $(n-1)$-dimensional submanifold of M (possibly empty).

Proof. This is Lemma 3.2 of Chapter 2 of the companion volume (Milnor's Lemma 3 there), applied to $g = a - f$, for which 0 is a regular value; the statement about $f^{-1}(a)$ is Lemma 2.1 of the same chapter. In the present setting the proof reads: near q with $f(q) < a$, the open set $f^{-1}(-\infty, a)$ lies in M^a; near q with $f(q) = a$, choose a chart $(x^1, \dots, x^n)$ with $\partial f/\partial x^n(q) \neq 0$. Then $y = (x^1, \dots, x^{n-1}, a - f)$ has Jacobian determinant $-\partial f/\partial x^n(q) \neq 0$ at q, so by the inverse function theorem (companion volume, Chapter 1, Theorem 4.1) it is a coordinate system on a neighbourhood W of q, in which $M^a \cap W = \{y^n \geq 0\}$ and $f^{-1}(a) \cap W = \{y^n = 0\}$. □

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3. The Hessian, the index and the nullity (Milnor pp. 4–5)

Let p be a critical point of f. For $v, w \in TM_p$ choose vector fields $\tilde{v}, \tilde{w}$ defined near p with $\tilde{v}_p = v$, $\tilde{w}_p = w$ (in a chart, if $v = \sum a_i\, \partial/\partial x^i|_p$, the constant-coefficient field $\sum a_i\, \partial/\partial x^i$ will do). Milnor defines the *Hessian*

$$f_{**}(v, w) = \tilde{v}_p(\tilde{w}(f)),$$

where $\tilde{w}(f)$ is the directional derivative of f along $\tilde{w}$, a smooth function near p, and $\tilde{v}_p(\cdot) = v(\cdot)$ is its derivative in the direction v.

Lemma B. f_{**} is a well-defined symmetric bilinear form on TM_p. In a chart, if $v = \sum a_i\, \partial/\partial x^i|_p$ and $w = \sum b_j\, \partial/\partial x^j|_p$, then

$$f_{**}(v, w) = \sum_{i,j} a_i b_j \frac{\partial^2 f}{\partial x^i \partial x^j}(p),$$

so the matrix $(\partial^2 f/\partial x^i \partial x^j(p))$ represents f_{**} in the basis $\partial/\partial x^1|_p, \dots, \partial/\partial x^n|_p$.

Proof. Milnor's argument: $\tilde{v}_p(\tilde{w}(f)) - \tilde{w}_p(\tilde{v}(f)) = [\tilde{v}, \tilde{w}]_p(f) = 0$, because the bracket $[\tilde{v}, \tilde{w}]_p$ is a tangent vector at p and f_* vanishes at p. So the expression is symmetric; it depends on $\tilde{v}$ only through $\tilde{v}_p = v$, and, being equal to $\tilde{w}_p(\tilde{v}(f))$, on $\tilde{w}$ only through w. The coordinate computation gives all of this at once. Write $\tilde{v} = \sum \alpha_i \partial_i$ and $\tilde{w} = \sum \beta_j \partial_j$ with smooth coefficients, $\alpha_i(p) = a_i$, $\beta_j(p) = b_j$. Then

$$\tilde{v}(\tilde{w}(f)) = \sum_{i,j} \alpha_i \, (\partial_i \beta_j) \, \partial_j f + \sum_{i,j} \alpha_i \beta_j \, \partial_i \partial_j f.$$

At p every $\partial_j f$ vanishes, so $\tilde{v}_p(\tilde{w}(f)) = \sum a_i b_j \, \partial_i \partial_j f(p)$. This depends only on v, w, is bilinear, and is symmetric because mixed partial derivatives commute. □

The vanishing of the first sum is the only place where criticality enters: at a non-critical point the same recipe depends on the extensions, and there is no invariant Hessian.

Definition. A critical point p is *non-degenerate* if the matrix $(\partial^2 f/\partial x^i \partial x^j(p))$ is non-singular. This does not depend on the chart: by Lemma B two charts give matrices S and S' representing the same form in two bases, so $S' = J^{\mathsf{T}} S J$ with J the invertible change-of-basis matrix. (The direct check Milnor alludes to is the chain rule $\frac{\partial^2 f}{\partial y^k \partial y^l} = \sum_{i,j} \frac{\partial x^i}{\partial y^k} \frac{\partial x^j}{\partial y^l} \frac{\partial^2 f}{\partial x^i \partial x^j} + \sum_i \frac{\partial^2 x^i}{\partial y^k \partial y^l} \frac{\partial f}{\partial x^i}$, whose last sum vanishes at a critical point.)

Definition. The *index* of a symmetric bilinear form H on a vector space V is the maximal dimension of a subspace of V on which H is negative definite. The *nullity* is the dimension of the *null space* $\{v \in V : H(v, w) = 0 \text{ for all } w \in V\}$. The index of f_{**} on TM_p is called the *index of f at p*.

If S represents H in a basis, v lies in the null space exactly when its coordinate vector lies in the kernel of S. Hence p is a non-degenerate critical point if and only if f_{**} has nullity 0. To compute indices we need one piece of linear algebra, which Milnor uses inside the proof of Lemma 2.2.

Lemma C. Let H be a symmetric bilinear form on an n-dimensional real vector space V, and suppose $V = A \oplus B$ where H is negative definite on A and $H(b, b) \geq 0$ for all $b \in B$. Then the index of H is $\dim A$. Consequently, if H has the diagonal matrix $\operatorname{diag}(d_1, \dots, d_n)$ in some basis, its index is $\#\{i : d_i < 0\}$ and its nullity is $\#\{i : d_i = 0\}$; and for any matrix S representing H, the index and the nullity are the numbers of negative and of zero eigenvalues of S, counted with multiplicity.

Proof. The index is at least $\dim A$ by definition. If L were a subspace with $\dim L > \dim A$ on which H is negative definite, then $\dim L + \dim B > n$, so $L \cap B$ contains some $v \neq 0$, and $H(v, v) < 0 \leq H(v, v)$, a contradiction. For a diagonal matrix take A spanned by the basis vectors with $d_i < 0$ and B by the others. The null space consists of the $\sum c_i e_i$ with $c_j d_j = H(\sum c_i e_i, e_j) = 0$ for all j, i.e. it is spanned by the e_j with $d_j = 0$. Finally, by the spectral theorem $S = Q D Q^{\mathsf{T}}$ with Q orthogonal and D diagonal with the eigenvalues of S on the diagonal; in the basis formed by the columns of Q (read as coordinate vectors) H has matrix $Q^{\mathsf{T}} S Q = D$. □

The torus, computed. With $f(u, v) = R + \rho + (R + \rho \cos v) \sin u$,

$$f_u = (R + \rho \cos v) \cos u, \qquad f_v = -\rho \sin v \sin u.$$

Since $R + \rho \cos v \geq R - \rho > 0$, $f_u = 0$ forces $\cos u = 0$, so $\sin u = \pm 1$, and then $f_v = 0$ forces $\sin v = 0$: the critical points are exactly p, q, r, s. The second derivatives are $f_{uu} =$

$-(R+\rho\cos v)\sin u$, $f_{uv} = -\rho\sin v\cos u$ (zero at the critical points) and $f_{vv} = -\rho\cos v\sin u$. The Hessian matrices are

$$p:\ \mathrm{diag}(R+\rho,\ \rho),\quad q:\ \mathrm{diag}(R-\rho,\ -\rho),\quad r:\ \mathrm{diag}(-(R-\rho),\ \rho),\quad s:\ \mathrm{diag}(-(R+\rho),\ -\rho).$$

All four are non-degenerate, of indices $0, 1, 1, 2$: the dimensions of the cells in §1.

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4. Hadamard's lemma and the lemma of Morse (Milnor pp. 5–8)

Lemma 2.1 (Hadamard). *(checked by machine: V1.7, in a stronger form)*[1] If f is C^∞ on a convex neighbourhood V of 0 in $\mathbb{R}^n$ and $f(0) = 0$, then there are C^∞ functions $g_1, \dots, g_n$ on V with $g_i(0) = \dfrac{\partial f}{\partial x_i}(0)$ and

$$f(x_1, \dots, x_n) = \sum_{i=1}^{n} x_i\, g_i(x_1, \dots, x_n) \qquad \text{on } V.$$

This is the Hadamard lemma of the companion volume (Chapter 6, proof of Lemma 1.1, for maps defined on all of $\mathbb{R}^m$). We need it on a convex neighbourhood, and the proof is two lines, so here it is.

Proof. For $x \in V$ the segment $\{tx : 0 \le t \le 1\}$ lies in V. The function $t \mapsto f(tx)$ is smooth on $[0,1]$ with derivative $\sum_i x_i\, \partial_i f(tx)$, so by the fundamental theorem of calculus

$$f(x) = f(x) - f(0) = \int_0^1 \frac{df(tx)}{dt}\, dt = \sum_{i=1}^{n} x_i \int_0^1 \frac{\partial f}{\partial x_i}(tx)\, dt.$$

Put $g_i(x) = \int_0^1 \partial_i f(tx)\, dt$; then $g_i(0) = \int_0^1 \partial_i f(0)\, dt = \partial_i f(0)$. The integrand is a C^∞ function of (t, x) on the open set $\{(t,x) : tx \in V\} \supseteq [0,1] \times V$. By the theorem on differentiation under the integral sign (on a compact neighbourhood K of a point of V, each x-derivative of the integrand is uniformly continuous on $[0,1] \times K$, which lets the difference quotients pass under the integral; induct on the order), g_i is C^∞ with $\partial^\alpha g_i(x) = \int_0^1 \partial_x^\alpha [\partial_i f(tx)]\, dt$. □

Only the segments from 0 were used: "star-shaped about 0" suffices for "convex".

Lemma 2.2 (Lemma of Morse). *(checked by machine: V1.8, in a stronger form)* If p is a non-degenerate critical point of f, of index λ, then p has a neighbourhood U with coordinates $(y^1, \dots, y^n)$, all vanishing at p, in which

$$f = f(p) - (y^1)^2 - \cdots - (y^\lambda)^2 + (y^{\lambda+1})^2 + \cdots + (y^n)^2$$

at every point of U.

[1] Checked by machine: the result has been verified in the Lean proof assistant. The numbers are those of the verified edition and of the comparison, which give the checked statement in full. *As stated*: the same statement. *Restated*: an equivalent statement in other terms. *With its hypotheses made explicit*: a hypothesis the book uses silently is stated. *In a stronger form*: the checked statement says more than the book's. *In a narrower form*: it covers less, a special setting or a weaker conclusion. *With a sign corrected*: the book's statement has a slip. *Not checked*: no part of the result has been verified. The section "Checked by machine" at the end of the chapter says which forms were checked, and covers the results that carry no such note.

Proof. **Part 1: in any such expression, λ is the index.** Suppose coordinates $(z^1, \dots, z^n)$ centred at p satisfy $f(q) = f(p) - \sum_{i\le\lambda}(z^i(q))^2 + \sum_{i>\lambda}(z^i(q))^2$. Then

$$\frac{\partial^2 f}{\partial z^i \partial z^j}(p) = \begin{cases} -2 & i = j \le \lambda, \\ 2 & i = j > \lambda, \\ 0 & \text{otherwise,} \end{cases}$$

so by Lemma B the matrix of f_{**} in the basis $\partial/\partial z^1|_p, \dots, \partial/\partial z^n|_p$ is $\operatorname{diag}(-2, \dots, -2, 2, \dots, 2)$ with λ entries -2. By Lemma C (this is Milnor's argument: a negative definite subspace of dimension greater than λ would meet the positive definite span of the last $n-\lambda$ basis vectors), λ is the index of f_{**}.

Part 2: existence. *Step 1 (reduction).* Take a chart around p, translate it so that p becomes $0 \in \mathbb{R}^n$, restrict it to an open ball U_0 about 0, and replace f by $f - f(p)$. So f is smooth on U_0, $f(0) = 0$, and 0 is a non-degenerate critical point.

Step 2 (a quadratic form with variable coefficients). By Lemma 2.1, $f = \sum_j x_j g_j$ on U_0 with $g_j(0) = \partial_j f(0) = 0$. Applying Lemma 2.1 to each g_j gives $g_j = \sum_i x_i h_{ij}$ with smooth h_{ij}, so $f = \sum_{i,j} x_i x_j h_{ij}$. Put $H_{ij} = \frac{1}{2}(h_{ij} + h_{ji})$; then $H_{ij} = H_{ji}$ and $f = \sum_{i,j} x_i x_j H_{ij}$. Differentiating twice at 0: in $\partial_k\partial_l(x_i x_j H_{ij})(0)$ every term in which a factor x_i or x_j survives undifferentiated vanishes, leaving $(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})H_{ij}(0)$. Summing, $\partial_k\partial_l f(0) = 2H_{kl}(0)$. So $(H_{ij}(0)) = \frac{1}{2}(\partial^2 f/\partial x_i \partial x_j(0))$ is non-singular. (Writing out the two integrals and changing variables shows $h_{ij}(x) = \int_0^1 (1-t)\,\partial_i\partial_j f(tx)\,dt$, already symmetric; we will not need this, but the interactive figure uses it.)

Step 3 (the induction). For $r = 1, \dots, n+1$ consider the statement:

> (P_r) On some neighbourhood U_r of 0 there are coordinates $u_1, \dots, u_n$ with $u(0) = 0$, signs $\epsilon_1, \dots, \epsilon_{r-1} \in \{\pm 1\}$, and smooth functions $H_{ij} = H_{ji}$ $(i, j \ge r)$ on U_r, such that
>
> $$f = \epsilon_1 u_1^2 + \cdots + \epsilon_{r-1} u_{r-1}^2 + \sum_{i,j\ge r} u_i u_j H_{ij} \quad \text{throughout } U_r,$$
>
> and the matrix $\big(H_{ij}(0)\big)_{i,j\ge r}$ is non-singular.

Step 2 is (P_1). We show $(P_r) \Rightarrow (P_{r+1})$ for $r \le n$.

(a) *Making $H_{rr}(0) \neq 0$.* An invertible linear substitution $u_{\ge r} = T w_{\ge r}$ of the last $n-r+1$ coordinates keeps the form of (P_r): $\sum_{i,j\ge r} u_i u_j H_{ij} = \sum_{a,b\ge r} w_a w_b \tilde{H}_{ab}$ with $\tilde{H} = T^{\mathsf{T}} H T$, still symmetric and smooth, with $\tilde{H}(0) = T^{\mathsf{T}} H(0) T$ non-singular, and $w(0) = 0$. The block $\big(H_{ij}(0)\big)_{i,j\ge r}$ is non-singular, hence not zero. If some diagonal entry $H_{kk}(0)$, $k \ge r$, is nonzero, interchange u_r and u_k. Otherwise pick $k \neq l$ (both $\ge r$) with $H_{kl}(0) \neq 0$, first interchange so that $k = r$, then substitute $u_r = w_r + w_l$, $u_l = w_r - w_l$: the new entry is $\tilde{H}_{rr}(0) = H_{rr}(0) + 2H_{rl}(0) + H_{ll}(0) = 2H_{rl}(0) \neq 0$.

(b) *Completing the square.* Let $\sigma \in \{\pm 1\}$ be the sign of $H_{rr}(0)$. By continuity there is a neighbourhood $U' \subseteq U_r$ of 0 on which H_{rr} has the sign σ at every point; put $g = \sqrt{|H_{rr}|} = \sqrt{\sigma H_{rr}}$,

smooth and positive on U'. Define new functions

$$v_i = u_i \ (i \neq r), \qquad v_r = g\left[u_r + \sum_{i>r} u_i \, H_{ir}/H_{rr}\right].$$

(c) *They are coordinates.* $v(0) = 0$. At 0, every term of v_r is a product with a factor u_i vanishing at 0, so $\partial v_r/\partial u_j(0)$ comes only from differentiating that factor: it is 0 for $j < r$, $g(0)$ for $j = r$, and $g(0)H_{jr}(0)/H_{rr}(0)$ for $j > r$. The Jacobian matrix of v with respect to u at 0 is therefore the identity except for row r, which is $(0, \dots, 0, g(0), *, \dots, *)$: it is upper triangular with determinant $g(0) \neq 0$. By the inverse function theorem (companion volume, Chapter 1, §1), $v_1, \dots, v_n$ are coordinates on some neighbourhood $U_{r+1} \subseteq U'$ of 0.

(d) *The identity.* Since $\sigma g^2 = H_{rr}$,

$$\sigma v_r^2 = H_{rr}u_r^2 + 2u_r \sum_{i>r} u_i H_{ir} + \frac{\left(\sum_{i>r} u_i H_{ir}\right)^2}{H_{rr}},$$

and therefore

$$\sum_{i,j\geq r} u_i u_j H_{ij} = \sigma v_r^2 + \sum_{i,j>r} u_i u_j H'_{ij}, \qquad H'_{ij} = H_{ij} - \frac{H_{ir}H_{jr}}{H_{rr}}.$$

As $u_i = v_i$ for $i \neq r$, this is $f = \sum_{i\leq r} \epsilon_i v_i^2 + \sum_{i,j>r} v_i v_j H'_{ij}$ with $\epsilon_r = \sigma$, the H'_{ij} smooth and symmetric.

(e) *Non-singularity persists.* $\left(H'_{ij}(0)\right)_{i,j>r}$ is the Schur complement of the entry $H_{rr}(0)$ in $\left(H_{ij}(0)\right)_{i,j\geq r}$, and $\det\left(H_{ij}(0)\right)_{i,j\geq r} = H_{rr}(0)\det\left(H'_{ij}(0)\right)_{i,j>r}$ (subtract multiples of the first row to clear the first column). So (P_{r+1}) holds.

Step 4 (conclusion). (P_{n+1}) says $f = \sum_{i=1}^{n} \epsilon_i v_i^2$ near 0. Renumber the coordinates so that the negative signs come first and call them $y^1, \dots, y^n$; restoring the constant and the original chart, $f = f(p) - \sum_{i\leq\lambda'}(y^i)^2 + \sum_{i>\lambda'}(y^i)^2$ near p. By Part 1, $\lambda' = \lambda$. □

> **Milnor's text skips…** the reason why, "after a linear change in the last $n - r + 1$ coordinates", one may assume $H_{rr}(0) \neq 0$. It needs the block $\left(H_{ij}(0)\right)_{i,j\geq r}$ to be nonzero, which is why the induction must carry its non-singularity along (step (e)); and when all its diagonal entries vanish, a coordinate interchange is not enough, and the substitution $u_r = w_r + w_l$, $u_l = w_r - w_l$ is needed. Also unstated: each new coordinate system must again vanish at 0, which the completed square does, and the Jacobian in (c) is triangular.

Corollary 2.3. *(checked by machine: V1.10, in a stronger form)* Non-degenerate critical points are isolated.

Proof. In the coordinates of Lemma 2.2, $df = \sum_i \pm 2y^i \, dy^i$, which vanishes only where all $y^i = 0$, that is, at p. So p is the only critical point in U. □

An interactive look at the lemma of Morse

The figure shows the contour lines of

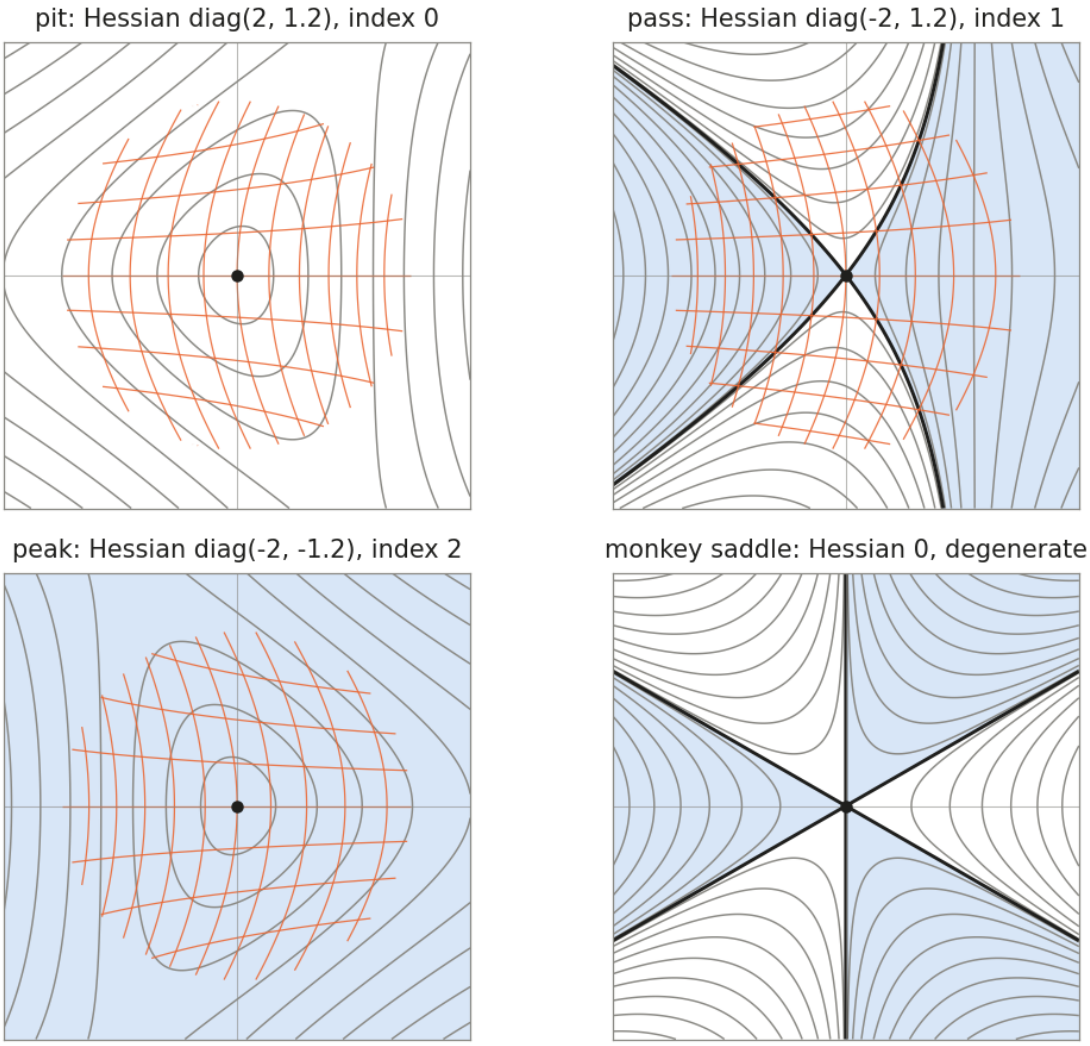
$$f(x, y) = a\,x^2 + b\,y^2 + \gamma\,(x^3 - 3xy^2)$$

near the critical point at the origin, with the region $\{f < 0\}$ shaded. The Hessian matrix at 0 is $\mathrm{diag}(2a, 2b)$; the panel reports the index and the nullity. When a and b are both nonzero, the figure also draws the Morse coordinates (y^1, y^2) produced by the proof above, as a curved grid: here $H_{11} = a + \gamma x$, $H_{12} = -\gamma y$, $H_{22} = b - \gamma x$, and step (b) gives $y^1 = \sqrt{|H_{11}|}\,(x + yH_{12}/H_{11})$, $y^2 = \sqrt{|D/H_{11}|}\;y$ with $D = H_{11}H_{22} - H_{12}^2$ (when $|b| > |a|$ the figure takes H_{22} as the pivot instead, exchanging the roles of x and y). Wherever the grid is drawn, the contour lines of f are exactly the circles or hyperbolas of $\pm(y^1)^2 \pm (y^2)^2$ in it; the cubic term only bends the grid.

Drag a and b through zero to move between a pit (index 0), a pass (index 1) and a peak (index 2). Shrink the window (slider R): for a non-degenerate point the picture converges to that of the quadratic part. Set $a = b = 0$ and the cubic term wins: the contours form Milnor's monkey saddle, with three valleys, which no change of coordinates can turn into a quadratic form.

(In the printed version this figure is a still image; the interactive version is in the web edition.)

$f = ax^2 + by^2 + \gamma(x^3 - 3xy^2)$: contours of f (grey), $\{f < 0\}$ (blue),
Morse coordinates (y^1, y^2) from the proof of Lemma 2.2 (orange grid)



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5. Degenerate critical points (Milnor pp. 8–9)

Milnor's examples, with the verifications he leaves to the pictures:

(a) $f(x) = x^3$ on $\mathbb{R}$: $f'(0) = f''(0) = 0$, so 0 is a degenerate critical point. It is isolated: $f'(x) = 3x^2 \neq 0$ for $x \neq 0$. Degenerate points need not be extrema.

(b) $F(x) = e^{-1/x^2} \sin^2(1/x)$ for $x \neq 0$, $F(0) = 0$. Every derivative of F on $x \neq 0$ is a finite sum of terms (polynomial in $1/x$)$\cdot e^{-1/x^2}\cdot$(product of sin and cos of $1/x$), bounded by $C|x|^{-N}e^{-1/x^2} \to 0$ as $x \to 0$; by induction and the mean value theorem F is C^∞ with all derivatives 0 at 0. So 0 is a degenerate critical point. It is not isolated: $F \geq 0$ and

$F(1/k\pi) = 0$, so each $1/k\pi$ is a minimum, hence critical, and $1/k\pi \to 0$. (By Corollary 2.3, a non-isolated critical point is necessarily degenerate.)

(c) $f(x,y) = x^3 - 3xy^2 = \operatorname{Re}(x+iy)^3$, the *monkey saddle*: $\nabla f = (3x^2 - 3y^2, -6xy)$ vanishes only at 0, and all second derivatives $6x, -6y, -6x$ vanish there. An isolated degenerate critical point, with three descending valleys.

(d) $f(x,y) = x^2$: $\nabla f = (2x, 0)$ vanishes exactly on the line $x = 0$, where the Hessian is $\operatorname{diag}(2,0)$, of nullity 1. Every critical point is degenerate, and the critical set is a submanifold of $\mathbb{R}^2$. (Milnor calls it "the x axis"; for $f = x^2$ it is the y axis.)

(e) $f(x,y) = x^2y^2$: $\nabla f = (2xy^2, 2x^2y)$ vanishes exactly where $xy = 0$, the union of the two axes. The Hessian $\begin{pmatrix} 2y^2 & 4xy \\ 4xy & 2x^2 \end{pmatrix}$ has determinant $4x^2y^2 - 16x^2y^2 = -12x^2y^2 = 0$ there, so every critical point is degenerate. The critical set is not a submanifold: it is neither discrete nor open in $\mathbb{R}^2$, so near the origin it would have to be a curve; but removing the origin from a small connected neighbourhood of it in the cross leaves four components, while removing a point from an arc leaves two.

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6. One-parameter groups of diffeomorphisms (Milnor pp. 9–11)

Definition. A *1-parameter group of diffeomorphisms* of M is a C^∞ map $\varphi : \mathbb{R} \times M \to M$ such that

1. for each $t \in \mathbb{R}$ the map $\varphi_t : M \to M$, $\varphi_t(q) = \varphi(t,q)$, is a diffeomorphism of M onto itself;
2. for all $t, s \in \mathbb{R}$, $\varphi_{t+s} = \varphi_t \circ \varphi_s$.

(Then $\varphi_0 = \mathrm{id}$, since $\varphi_0 \circ \varphi_0 = \varphi_0$ and φ_0 is bijective.)

For a smooth curve $t \mapsto c(t) \in M$ the *velocity vector* $\frac{dc}{dt} \in TM_{c(t)}$ is defined by $\frac{dc}{dt}(f) = \lim_{h\to 0} \frac{f(c(t+h)) - f(c(t))}{h}$; in coordinates $\frac{dc}{dt} = \sum_i \frac{d(x^i \circ c)}{dt} \frac{\partial}{\partial x^i}$. Given a 1-parameter group φ, the vector field X with

$$X_q(f) = \lim_{h\to 0} \frac{f(\varphi_h(q)) - f(q)}{h}$$

is said to *generate* φ. It is smooth: its components $X^i(q) = \partial_t (x^i \circ \varphi)(0,q)$ are smooth because φ is. For each fixed q the curve $t \mapsto \varphi_t(q)$ satisfies

$$\frac{d\varphi_t(q)}{dt} = X_{\varphi_t(q)}, \qquad \varphi_0(q) = q,$$

because, with $p = \varphi_t(q)$ and $\varphi_{t+h} = \varphi_h \circ \varphi_t$,

$$\frac{d\varphi_t(q)}{dt}(f) = \lim_{h\to 0} \frac{f(\varphi_{t+h}(q)) - f(\varphi_t(q))}{h} = \lim_{h\to 0} \frac{f(\varphi_h(p)) - f(p)}{h} = X_p(f).$$

A curve c with $\frac{dc}{dt} = X_{c(t)}$ for all t in an interval is an *integral curve* of X. In local coordinates $u^1, \dots, u^n$, with $X = \sum X^i \partial/\partial u^i$, this is the familiar system $du^i/dt = X^i(u^1, \dots, u^n)$, $i = 1, \dots, n$. What Milnor calls "well known" is the following.

Theorem D (ordinary differential equations). Let $W \subseteq \mathbb{R}^n$ be open and $F : W \to \mathbb{R}^n$ smooth.

(a) *Uniqueness.* If $u_1, u_2 : J \to W$ are differentiable solutions of $u' = F(u)$ on an interval J (open, closed or half-open) and $u_1(t_0) = u_2(t_0)$ for one $t_0 \in J$, then $u_1 = u_2$ on J.

(b) *Existence and smooth dependence.* Every $x_0 \in W$ has an open neighbourhood $W_0 \subseteq W$, and there are $\epsilon > 0$ and a C^∞ map $\Phi : (-\epsilon, \epsilon) \times W_0 \to W$, such that $\partial_t \Phi(t, x) = F(\Phi(t, x))$ and $\Phi(0, x) = x$.

Part (b) is the Picard–Lindelöf theorem together with the smooth dependence of solutions on initial conditions; Milnor cites Graves, *The Theory of Functions of Real Variables*, p. 166. Complete proofs are in Lang, *Fundamentals of Differential Geometry*, Ch. IV, §1, and in Lee, *Introduction to Smooth Manifolds* (2nd ed.), Appendix D. Part (a) is short:

Proof of (a). The set $S = \{t \in J : u_1(t) = u_2(t)\}$ contains t_0 and is closed in J. It is also open in J. Let $s \in S$ and let $B \subseteq W$ be a closed ball about $u_1(s)$. On B, F is Lipschitz with constant $L = \max_B \|DF\|$ (mean value inequality on the convex set B). By continuity there is $\delta > 0$ with $L\delta \le \frac{1}{2}$ and $u_1(t), u_2(t) \in B$ for $t \in I = J \cap [s - \delta, s + \delta]$. Let $m = \sup_I \|u_1 - u_2\|$, finite as both curves stay in B (I need not be compact when J is open). For $t \in I$,

$$\|u_1(t) - u_2(t)\| = \Big\| \int_s^t (F(u_1) - F(u_2)) \Big\| \le L\,|t - s|\, m \le \tfrac{1}{2} m,$$

so $m \le \frac{1}{2}m$, $m = 0$, and $I \subseteq S$. An interval is connected, so $S = J$. □

Lemma E (uniqueness on a manifold). Let X be a smooth vector field on the Hausdorff manifold M. Two integral curves $\gamma_1, \gamma_2 : J \to M$ of X on the same interval J (of any kind) that agree at one time agree on all of J.

Proof. $S = \{t \in J : \gamma_1(t) = \gamma_2(t)\}$ is nonempty. It is closed in J because M is Hausdorff (the diagonal of $M \times M$ is closed, and S is its preimage under $t \mapsto (\gamma_1(t), \gamma_2(t))$). It is open in J: if $s \in S$, take a chart around $\gamma_1(s)$; for $t \in J$ near s both curves lie in the chart and solve the same coordinate system of equations with the same value at s, so they agree near s by Theorem D(a). Hence $S = J$. □

> **A hidden hypothesis.** The Hausdorff condition is needed. On the line with two origins, the field d/dx has two different integral curves starting at the point -1: they agree except at the time $t = 1$, when each passes through a different origin.

Lemma 2.4. *(checked by machine: V1.15, in a narrower form)* Every smooth vector field on M that vanishes outside some compact set $K \subset M$ generates a 1-parameter group of diffeomorphisms of M, and only one.

Proof. Step 1 (local flows). For each $x \in M$, Theorem D(b) applied in a chart around x gives an open neighbourhood U_x of x, a number $\epsilon_x > 0$ and a smooth map $\Phi^x : (-\epsilon_x, \epsilon_x) \times U_x \to M$ such that $t \mapsto \Phi^x(t, q)$ is an integral curve of X starting at q.

Step 2 (a uniform time). K is covered by finitely many of the U_x, say $U_{x_1}, \dots, U_{x_m}$; let $\epsilon_0 = \min_k \epsilon_{x_k} > 0$. For $|t| < \epsilon_0$ define $\varphi(t, q) = \Phi^{x_k}(t, q)$ if $q \in U_{x_k}$, and $\varphi(t, q) = q$ if $q \notin K$. This is consistent: if $q \notin K$ then $X_q = 0$, so the constant curve is an integral curve, and any two integral curves on $(-\epsilon_0, \epsilon_0)$ starting at q coincide by Lemma E. It is smooth on $(-\epsilon_0, \epsilon_0) \times M$, because it equals Φ^{x_k} on $(-\epsilon_0, \epsilon_0) \times U_{x_k}$ and the projection to M on $(-\epsilon_0, \epsilon_0) \times (M \setminus K)$; the set $M \setminus K$ is open since K is compact and M is Hausdorff, and these open sets cover.

Step 3 (the group law for small times). If $|s|, |t|, |s+t| < \epsilon_0$, then $\varphi_{t+s} = \varphi_t \circ \varphi_s$. Indeed, for fixed s and q, the curves $\tau \mapsto \varphi_{\tau+s}(q)$ and $\tau \mapsto \varphi_\tau(\varphi_s(q))$ are integral curves on the interval $\{\tau : |\tau| < \epsilon_0, |\tau + s| < \epsilon_0\}$, which contains 0 and t, with the same value $\varphi_s(q)$ at $\tau = 0$; Lemma E. In particular $\varphi_t \circ \varphi_{-t} = \varphi_0 = \mathrm{id}$, so each φ_t, $|t| < \epsilon_0$, is a diffeomorphism.

Step 4 (all times: Milnor's definition). Put $\delta = \epsilon_0/2$ and $\psi = \varphi_\delta$, a diffeomorphism with inverse $\varphi_{-\delta}$. Every $t \in \mathbb{R}$ can be written as $t = k\delta + r$ with $k \in \mathbb{Z}$ and $|r| < \delta$; set

$$\varphi_t = \psi^k \circ \varphi_r,$$

where ψ^k is the k-fold composite of ψ (of $\psi^{-1} = \varphi_{-\delta}$ if $k < 0$).

Well defined. The decomposition is not unique ($\delta/2 = 0 \cdot \delta + \delta/2 = 1 \cdot \delta - \delta/2$). If $t = k\delta + r = k'\delta + r'$, then $|k - k'|\delta = |r' - r| < 2\delta$, so $k' \in \{k-1, k, k+1\}$; if $k' = k+1$ then $r = r' + \delta$ and Step 3 gives $\varphi_r = \psi \circ \varphi_{r'}$, whence $\psi^k \circ \varphi_r = \psi^{k+1} \circ \varphi_{r'}$. The case $k' = k - 1$ is symmetric. By Step 3 the new φ_t agrees with the old one for $|t| < \epsilon_0$ (if $\delta \le t < \epsilon_0$, $\psi \circ \varphi_{t-\delta} = \varphi_t$).

Smooth. Given $t_0 = k\delta + r_0$ with $|r_0| < \delta$, for $|t - t_0| < \delta - |r_0|$ we may use the same k, so $\varphi(t, q) = \psi^k(\varphi(t - k\delta, q))$ there: a composite of smooth maps.

Integral curves. For $|u| < \delta$, Step 3 gives $\psi \circ \varphi_u = \varphi_{\delta+u} = \varphi_u \circ \psi$, and likewise for ψ^{-1}; so ψ^k commutes with φ_u. Hence, for $|t - t_0| < \delta - |r_0|$,

$$\varphi_t(q) = \psi^k(\varphi_{t-t_0}(\varphi_{r_0}(q))) = \varphi_{t-t_0}(\psi^k(\varphi_{r_0}(q))) = \varphi_{t-t_0}(\varphi_{t_0}(q)),$$

using Step 3 for the first equality. The right side, as a function of t, is an integral curve of X near t_0. So $t \mapsto \varphi_t(q)$ is an integral curve of X on all of $\mathbb{R}$.

Step 5 (the group law; uniqueness). For fixed s and q, both $t \mapsto \varphi_{t+s}(q)$ and $t \mapsto \varphi_t(\varphi_s(q))$ are integral curves on $\mathbb{R}$ with the same value at $t = 0$, so $\varphi_{t+s} = \varphi_t \circ \varphi_s$ by Lemma E. Then φ_{-t} is a smooth inverse of φ_t, so φ is a 1-parameter group, and its generator is X (the integral-curve property at $t = 0$). If φ' is another 1-parameter group generated by X, then each $t \mapsto \varphi'_t(q)$ is an integral curve starting at q, as computed above, so $\varphi' = \varphi$ by Lemma E. □

> **Milnor's text skips…** three verifications behind his "it is not difficult to verify" that φ_t is well defined, smooth, and satisfies $\varphi_{t+s} = \varphi_t \circ \varphi_s$: the decomposition $t = k(\epsilon_0/2) + r$ is not unique, so independence of the choice has to be checked; the group law for all times is most easily obtained by first showing that $t \mapsto \varphi_t(q)$ is an integral curve, which needs $\varphi_{\epsilon_0/2}$ to commute with the small φ_u; and the uniqueness of integral curves on a manifold uses the Hausdorff property.

Remark (Milnor p. 11). The hypothesis that X vanishes outside a compact set cannot be omitted. Let $M = (0, 1)$ and $X = d/dt$. If a 1-parameter group φ were generated by X, then for $q \in (0, 1)$ the curve $\gamma(t) = \varphi_t(q)$ would satisfy $\gamma'(t) = 1$ for all $t \in \mathbb{R}$, hence $\gamma(t) = q + t$, which leaves $(0, 1)$ at $t = 1 - q$. So X generates no 1-parameter group.

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7. Application: the four equilibria of a double pendulum

The model. Rods of lengths l_1, l_2, point masses m_1, m_2 at their tips, gravity g, angles θ_1, θ_2 from the downward vertical. The heights of the masses are $-l_1 \cos\theta_1$ and $-l_1 \cos\theta_1 - l_2 \cos\theta_2$, so the

potential energy is the smooth function on the torus

$$V(\theta_1,\theta_2) = -(m_1+m_2)\,g\,l_1\cos\theta_1 - m_2\,g\,l_2\cos\theta_2.$$

The kinetic energy is $T = \frac{1}{2}\dot\theta^{\mathsf T} K(\theta)\,\dot\theta$ with the mass matrix

$$K(\theta) = \begin{pmatrix} (m_1+m_2)l_1^2 & m_2 l_1 l_2\cos(\theta_1-\theta_2) \\ m_2 l_1 l_2\cos(\theta_1-\theta_2) & m_2 l_2^2 \end{pmatrix},$$

positive definite (its determinant is $m_2 l_1^2 l_2^2(m_1 + m_2\sin^2(\theta_1-\theta_2)) > 0$).

The critical points and their indices. $\partial V/\partial\theta_1 = (m_1+m_2)gl_1\sin\theta_1$ and $\partial V/\partial\theta_2 = m_2 g l_2\sin\theta_2$ vanish exactly when $\theta_1,\theta_2 \in \{0,\pi\}$: the four equilibria. The Hessian is diagonal, $\operatorname{diag}\big((m_1+m_2)gl_1\cos\theta_1,\ m_2 g l_2\cos\theta_2\big)$, so by Lemma C:

equilibrium (θ_1,θ_2)	V	Hessian signs	index
$(0,0)$, both down	$-(m_1+m_2)gl_1 - m_2 g l_2$	$(+,+)$	0
$(0,\pi)$, outer rod up	$-(m_1+m_2)gl_1 + m_2 g l_2$	$(+,-)$	1
$(\pi,0)$, inner rod up	$(m_1+m_2)gl_1 - m_2 g l_2$	$(-,+)$	1
(π,π), both up	$(m_1+m_2)gl_1 + m_2 g l_2$	$(-,-)$	2

All four are non-degenerate: V is a Morse function on the torus with one pit, two passes and one peak, the pattern of Milnor's torus.

The Morse charts, explicitly. Here the lemma of Morse can be seen with the naked eye. From $-\cos\theta = -1 + 2\sin^2(\theta/2) = 1 - 2\cos^2(\theta/2)$, put $k_1 = \sqrt{2(m_1+m_2)gl_1}$, $k_2 = \sqrt{2m_2 g l_2}$. Near the pass $(0,\pi)$, the functions $y^1 = k_1\sin(\theta_1/2)$ and $y^2 = k_2\cos(\theta_2/2)$ are coordinates on $\{|\theta_1| < \pi,\ 0 < \theta_2 < 2\pi\}$ (their Jacobian is $\operatorname{diag}(\frac{k_1}{2}\cos\frac{\theta_1}{2}, -\frac{k_2}{2}\sin\frac{\theta_2}{2})$, invertible there), and on that whole region

$$V = V(0,\pi) + (y^1)^2 - (y^2)^2.$$

Consequently, near the pass the set $\{V \le V(0,\pi)\}$ is $\{|y^1| \le |y^2|\}$: two opposite sectors, the outer rod tipped to the left or to the right, meeting only at the equilibrium. This is what a mountain pass is, and it is the local picture behind the attachment of a 1-cell in Chapter 2.

Small motions and the index. Linearizing Lagrange's equations at an equilibrium θ^* (the terms quadratic in $\dot\theta$ drop out) gives $K\ddot\xi + H\xi = 0$, with $K = K(\theta^*)$ and H the Hessian of V at θ^*. Since K is positive definite, $\langle a,b\rangle_K = a^{\mathsf T}Kb$ is an inner product, and $K^{-1}H$ is self-adjoint for it. By the spectral theorem there is a K-orthonormal basis e_1, e_2 with $He_i = \nu_i K e_i$; in it the form H has matrix $\operatorname{diag}(\nu_1,\nu_2)$, so by Lemma C

$$\text{index of } V \text{ at } \theta^* = \#\{i : \nu_i < 0\}.$$

Writing $\xi = c_1 e_1 + c_2 e_2$ decouples the equations into $\ddot c_i + \nu_i c_i = 0$: a mode with $\nu_i > 0$ oscillates with angular frequency $\sqrt{\nu_i}$; a mode with $\nu_i < 0$ has solutions $e^{\pm\sqrt{-\nu_i}\,t}$ and grows exponentially (its "frequency" $\sqrt{\nu_i}$ is imaginary). So the index is the number of unstable modes, whatever the masses and lengths. The ν_i are the roots of $\det(H - \nu K) = 0$.

Numbers. Take $m_1 = m_2 = 1$ kg, $l_1 = l_2 = 1$ m, $g = 9.81$ m/s^2, and write $G = g/l = 9.81$ s^{-2}. Dividing by ml^2, $K = \begin{pmatrix} 2 & \cos(\theta_1 - \theta_2) \\ \cos(\theta_1 - \theta_2) & 1 \end{pmatrix}$ and $H = \operatorname{diag}(2G\cos\theta_1, G\cos\theta_2)$.

- $(0,0)$: $\det(H - \nu K) = 2(G-\nu)^2 - \nu^2$, so $\nu = (2 \mp \sqrt{2})G = 5.747$ and 33.49 s^{-2}: two oscillations, $\omega = 2.397$ and 5.787 rad/s, periods 2.62 s and 1.09 s. (These are the textbook normal modes $\omega^2 = (2 \mp \sqrt{2})\, g/l$ of the equal double pendulum.)
- $(0,\pi)$ and $(\pi,0)$: now $\cos(\theta_1 - \theta_2) = -1$, and in both cases $\det(H - \nu K) = \nu^2 - 2G^2$, so $\nu = \pm\sqrt{2}\, G = \pm 13.87$ s^{-2}: one oscillation at 3.725 rad/s and one unstable mode growing like $e^{3.725\, t}$ (amplitude $\times e$ every 0.27 s). Index 1.
- (π,π): $\det(H - \nu K) = 2(G+\nu)^2 - \nu^2$, so $\nu = -(2 \mp \sqrt{2})G$: two unstable modes, growth rates 2.397 and 5.787 s^{-1}. Index 2.

In each case the count of negative ν agrees with the index in the table, as Lemma C guarantees. The same bookkeeping is standard in chemistry, where the potential energy of a nonlinear molecule of N atoms is a function of its $3N - 6$ internal coordinates: a stable conformer is a critical point of index 0, and a *transition state* is defined as a critical point of index 1, recognized in a normal-mode calculation by exactly one imaginary frequency. Its unique unstable direction is the reaction coordinate, and by the lemma of Morse the energy surface near it is, in suitable coordinates, exactly the saddle $E_0 - (y^1)^2 + (y^2)^2 + \cdots + (y^{3N-6})^2$.

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8. Summary of what was silently assumed, now made explicit

Where	Milnor says	What is actually needed
p. 1	(1) "If $a < 0 < f(p)$", as reproduced in the scan	$f(p) = 0$, as the torus touches V at p: the condition is $a < 0 = f(p)$
p. 2	e^k "with length ≤ 1" (the display in the scan reads $\|x\| < 1$)	the closed disk, so that $\dot{e}^k$ is its boundary
p. 4	"M^a is a smooth manifold-with-boundary"	companion volume, Chapter 2, Lemma 3.2, applied to $a - f$: the chart $(x^1, \ldots, x^{n-1}, a - f)$ (Proposition A)
p. 4	"v and w have extensions" to vector fields	local fields suffice; constant coefficients in a chart
p. 4	"It can be checked directly" that non-degeneracy does not depend on the chart	the second-order chain rule; its first-order term vanishes at a critical point (§3)
p. 5	"clearly well-defined"	criticality kills the first-order terms (Lemma B)
p. 5	Lemma 2.1 for "a convex neighborhood"	the Hadamard lemma of the companion volume (Chapter 6), on a domain star-shaped about 0; differentiation under the integral sign

Where	Milnor says	What is actually needed
p. 6	"a subspace … would intersect V, which is clearly impossible"	dimension count $\dim L + \dim B > n$ (Lemma C)
p. 7	"After a linear change in the last $n - r + 1$ coordinates" one may assume $H_{rr}(0) \neq 0$	non-singularity of the remaining block carried through the induction (Schur complement); the substitution $u_r = w_r + w_l$, $u_l = w_r - w_l$ when all diagonal entries vanish
p. 7	"$v_1, \dots, v_n$ will serve as coordinate functions"	the Jacobian at 0 is triangular with diagonal entry $g(0) \neq 0$
p. 9	(d) "the set of critical points … is the x axis"	for $f = x^2$ it is the y axis $\{x = 0\}$
p. 10	it is "well known" that the equation has, locally, a unique solution depending smoothly on the initial condition	Theorem D; uniqueness on a manifold needs M Hausdorff (Lemma E)
p. 11	"Setting $\varphi_t(q) = q$ for $q \notin K$ … smooth"	$M \setminus K$ is open because a compact set in a Hausdorff space is closed
p. 11	"It is not difficult to verify" that φ_t is well defined, smooth, and satisfies $\varphi_{t+s} = \varphi_t \circ \varphi_s$	independence of the decomposition $t = k\epsilon_0/2 + r$; commutation of $\varphi_{\epsilon_0/2}$ with small φ_u; group law via uniqueness of integral curves (Lemma 2.4, Steps 4–5)
p. 11	the Remark on $(0, 1)$	integral curves of d/dt are $q + t$, which leave $(0, 1)$
throughout	"manifold"	Hausdorff, second countable, without boundary

Checked by machine. Milnor's §2 has been verified in the Lean proof assistant. Lemma 2.1, examples (a), (c), (d) and (e), and Lemma 2.4 are checked in a finite-dimensional real vector space; Lemma B and the independence of the chart (only C^2 at p is used), the lemma of Morse with λ the index, and Corollary 2.3 are checked there and on an arbitrary smooth manifold, with no Hausdorff hypothesis. The lemma of Morse is proved by a single application of the inverse function theorem instead of the induction, and the smooth dependence on the initial point (the substance of Theorem D(b)) by the inverse function theorem on a space of curves rather than quoted. On the upright torus of §1 the height function is checked to have exactly four critical points, non-degenerate of indices $0, 1, 1, 2$, so that $C_0 - C_1 + C_2 = 0$ for every Morse function on it (Chapter 3). Not checked: the shapes of the torus's sublevel sets, example (b), Lemma 2.4 on a manifold, and the application; Proposition A is checked as part of the companion volume. See the verified edition.